

Malka Wickramarathna

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Education

University of Moratuwa 2025 – 2029 (Expected)
B.Sc. (Hons.) Electronic and Telecommunication Engineering

- Current GPA: **4.0/4.0**
- **Key Coursework:** Mathematical Analysis, ODEs and Vectors (A+), Electrical Fundamentals (A+), Programming Fundamentals (A+), Mechanics (A+), Fluid Mechanics (A+), Properties of Materials (A+)

Royal College, Colombo 2022 - 2024

- **G.C.E. Advanced Level 2023:** Ranked **5th in District** and **14th in Island** with Distinctions (A's) in Mathematics, Physics, Chemistry, and General English.

D.S. Senanayake College, Colombo 2015 - 2021

- **G.C.E. Ordinary Level 2020:** Achieved Distinctions in all 9 subjects (9A's).

Technical Projects

Electrocoagulation-based Household Water Purifier Prototype Jan 2026 - Jul 2026
Led the Group Project (Engineering Design Project) of 4 members

- Conceptualized a novel, affordable water purification system targeted at affected regions by excess amounts of Hardness, Fluoride, Heavy Metals, and Organic Pollutants in groundwater.
- Led the broad design of the water flow, physical placement of subsystems, and the structural rigidity of the system, considering expert advice and guidance.
- Built a proof-of-concept TRL-4 prototype and validated its functionality using synthetic groundwater in a laboratory setting.

Project Phoenix (CubeSatellite) Oct 2025 - Jan 2026
Electronic Club of UoM in collaboration with SEDS Mora

- Developed the Ground Station User Interface (UI) for a high-altitude weather balloon using **Python** and **HTML** for real-time telemetry visualization.
- Collaborated in the validation and testing of the **LoRa-based** communication sub-system.

Smart Hiking Pole Sept 2025 - Dec 2025
Group project of 4 members

- Designed an **ESP32-based** smart hiking pole integrating **GPS** and **IMU** sensors for location tracking in extreme environments.
- Developed a path-retracing algorithm utilizing logged GPS coordinates to generate return routes for users.
- Engineered a functional breadboard prototype, addressing technical challenges in signal acquisition and inte-

gration.

Skills

- **Programming & Scripting:** Python (Proficient), C/C++, LaTeX, HTML
- **Engineering Software:** MATLAB, Altium Designer
- **Languages:** English (Proficient), Sinhala (Native)

Selected Awards & Honors

G.C.E. Advanced Level Examination Achievements

2023

- **Presidential Appreciation Award:** Awarded by the President of Sri Lanka to the highest-ranking students in the country for demonstrating exceptional academic aptitude (District Rank 5).
- **Dr. J. Nanayakkara Memorial Prize:** Awarded for achieving the second-highest score in Physics within the 2023 Advanced Level cohort at Royal College, Colombo.
- **Cameron Samarasinghe Memorial Prize:** Awarded for outstanding performance in Combined Mathematics.
- **Royal College OBA (North America):**
 - **2nd Place in Physical Science Stream:** Awarded for outstanding performance in the Physical Science stream.
 - **3rd Place in Overall Performance:** Awarded for consistent academic excellence across all streams.

Professional Affiliations

IEEE Student Member

Since 2026

Interests

- Classical Piano
- Swimming